FinInsights

System Documentation

Enterprise-Grade Multi-Tenant Feature Intelligence & Growth Analytics

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Abstract

FinInsights provides a real-time, multi-tenant analytics operating layer for fintech products. It connects NexaBank telemetry to a low-latency OLAP stack and produces business-ready insights with governance controls.

1. Executive Summary & Project Scope

FinInsights is a high-throughput analytics system designed to convert raw behavioral telemetry into decision-grade intelligence for fintech organizations. By integrating the production-ready NexaBank platform with a real-time OLAP pipeline, FinInsights enables a single operational view across products, teams, and tenants.

The platform addresses Data Gravity and Batch Lag. Traditional BI systems often expose stale data, causing delayed response to conversion risks like drop-offs in KYC or payment completion. FinInsights uses a Lambda-Lite model to keep event-to-visibility latency typically under five seconds while preserving tenant-level isolation.

2. Technology Stack & Design Rationale

FinInsights follows the KFC architecture pattern (Kafka, FastAPI, ClickHouse), a proven blueprint for high-concurrency analytical systems.

Reading guide: this section explains what each component does, while later sections explain why the design remains low-latency, scalable, and operationally safe.

Component	Technology	Why It Was Chosen
Ingestion	FastAPI (Python 3.11)	Async IO enables high concurrency with low overhead and predictable latency.
Broker	Confluent Kafka 7.4	Decouples banking traffic from analytics processing and ensures durable event streams.
Analytics Database	ClickHouse 24.3 LTS	Columnar OLAP engine optimized for aggregation-heavy workloads at scale.
Frontend	Next.js 15	App Router and efficient rendering model support fast iterative product delivery.
Local AI	Ollama (Llama 3.2:1b)	On-prem inference enables data sovereignty for regulated fintech analytics.

3. Platform Capabilities and Feature Coverage

- **Feature Analytics:** usage trend chart, top feature leaderboard, and intensity heatmap.
- **Funnel Analysis:** stage conversion, leak detection, and optimization recommendations.
- **Governance and Security:** tracking toggles, operational controls, and auditable activity history.
- **License Intelligence:** licensed vs used features, underutilized premium modules, and upsell signals.
- **User Journey Mapping:** per-user event timeline for root-cause analysis.
- **Predictive + AI:** adoption forecasting and executive summary generation.
- **Multi-Tenant Comparison:** side-by-side performance for nexabank and safexbank.

4. Feature Taxonomy and Event Governance

FinInsights enforces canonical event naming to maintain consistency from ingestion to insight. Normalization and alias handling reduce duplicate metrics and preserve business meaning across apps.

Taxonomy Domain	Event Pattern	Representative Examples	Primary Insight
Authentication	login.*.*, register.*.*	login.auth.success, register.auth.success	Entry conversion and authentication friction
Navigation	*.page.view, *.dashboard.view	dashboard.page.view, account.page.view	Top journeys, discoverabil ity, feature awareness
Transactions	transaction.*.*, payee.*.*	transaction.pay_now.success, payee.create.success	Revenue-linked behavior and operational throughput
Loan Funnel	loan.*.*, kyc.*.*	loan.applied.success, loan.kyc_completed.success	KYC drop-off, completion velocity, risk hotspots
Pro Features	pro.*.*	pro.crypto-trading.page.view	License utilization, upsell opportunity, adoption depth

5. Real-Time Pipeline: Under-Five-Second Feedback Loop

- **Event Ingestion (<10ms typical):** NexaBank emits non-blocking background telemetry to the Ingestion API.
- **Streaming (<5ms append path):** Events are written to Kafka topics, decoupling producers from downstream consumers.
- **Batch Processing (~500ms windows):** Worker services consume in micro-batches, optimized for ClickHouse write efficiency.
- **Live Delivery (<100ms push path):** WebSocket updates deliver KPI deltas directly to active dashboard clients.

6. ClickHouse Data Model (feature_intelligence)

A. Raw Event Store: events_raw

The primary fact table stores immutable streaming telemetry. It uses MergeTree, monthly partitioning, and ordered keys beginning with tenant_id to maximize locality and bounded scans.

Column	Type	Description
tenant_id	String	Tenant boundary key used for strict logical isolation and scoped querying.
event_name	String	Canonical event taxonomy, for example login.auth.success.
user_id	String	An anonymized user identifier for behavior analysis.
timestamp	DateTime	Event occurrence time used for real-time and historical analytics.
metadata	String (JSON)	Context payload such as device, channel, location, and session hints.

B. Aggregates and System-State Tables

- **daily_feature_usage:** AggregatingMergeTree table fed by materialized view for chart-ready rollups.
- **tenant_licenses:** ReplacingMergeTree state table for plan and entitlement evolution.
- **tracking_toggles:** Dynamic telemetry control surface for feature-level governance.
- **config_audit_log:** Append-only compliance trail for high-trust operational changes.
- **ai_reports:** Cached narrative intelligence snapshots for low-latency executive summaries.

7. Feature Toggles, License Usage, and User Journey Mapping

A. Feature Toggle Control

- Feature toggles allow admins to enable or disable telemetry per mapped capability.
- Changes are recorded with actor and timestamp for operational traceability.
- Global toggle semantics support coherent behavior across tenant boundaries.

B. License Usage Intelligence

- Measures usage of entitled premium features against contract coverage.
- Flags underused licensed features and unlicensed but active capabilities.
- Supports revenue optimization and product packaging decisions.

C. User Journey Mapping

- Builds complete user-level event timelines from entry to conversion actions.
- Helps isolate friction points hidden behind aggregate metrics.
- Improves debugging and product experimentation confidence.

8. WebSocket Live Dashboarding

FinInsights removes the dependency on manual refresh. Connected dashboard sessions are grouped by tenant context. Each successful processor commit can trigger a KPI broadcast to relevant subscribers, enabling live awareness with reduced query churn.

Operational impact: compared with blind polling models, this approach reduces unnecessary read pressure and improves perceived responsiveness for analytical workflows.

9. Deployability and Scalability

Why It Is Deployable

- Containerized stack of coordinated services minimizes dependency mismatch and setup drift.

- Environment parity via configuration-driven deployment supports local, staging, and cloud targets without code forks.
- Automated bootstrapping initializes stream infrastructure and baseline resources at startup.

Why It Is Scalable

- Kafka partitions and worker replicas can be increased for horizontal throughput growth.
- ClickHouse can be expanded through sharding and distributed execution as data volume grows.
- Stateless ingestion and analytics APIs are load-balancer friendly for high request concurrency.

10. Multi-Tenant Isolation and RBAC

FinInsights enforces logical isolation by tenant_id at every query boundary. In this deployment, NexaBank serves two tenant contexts: its own tenant (**nexabank**) and an external partner tenant (**safexbank**). This structure enables clean cross-tenant benchmarking while preserving scoped access control.

- Tenant-scoped filtering is applied at row level in analytics storage and retrieval paths.
- Role-based access (RBAC) policies are dynamically configurable and support rapid permission updates.
- Administrative controls include governance toggles and auditable state transitions.

11. AI and Predictive Intelligence

FinInsights integrates local AI inference through Ollama, enabling private, context-aware narrative reporting without external data egress. The analytics layer supplies summarized metric signals, and the model transforms them into concise executive findings and directional recommendations.

- Data sovereignty is preserved by keeping sensitive metrics within controlled infrastructure.
- Narrative summaries accelerate decision-making for product, compliance, and leadership teams.
- Predictive surfaces complement descriptive dashboards with forward-looking trend signals.

12. Visual Product Walkthrough

The following screenshots capture the FinInsights analytical surfaces used by operations, product, growth, and leadership stakeholders.

Dashboard Overview

This page provides the operational command center: KPI snapshots, traffic and adoption trends, and tenant-scoped health indicators for rapid decision-making.

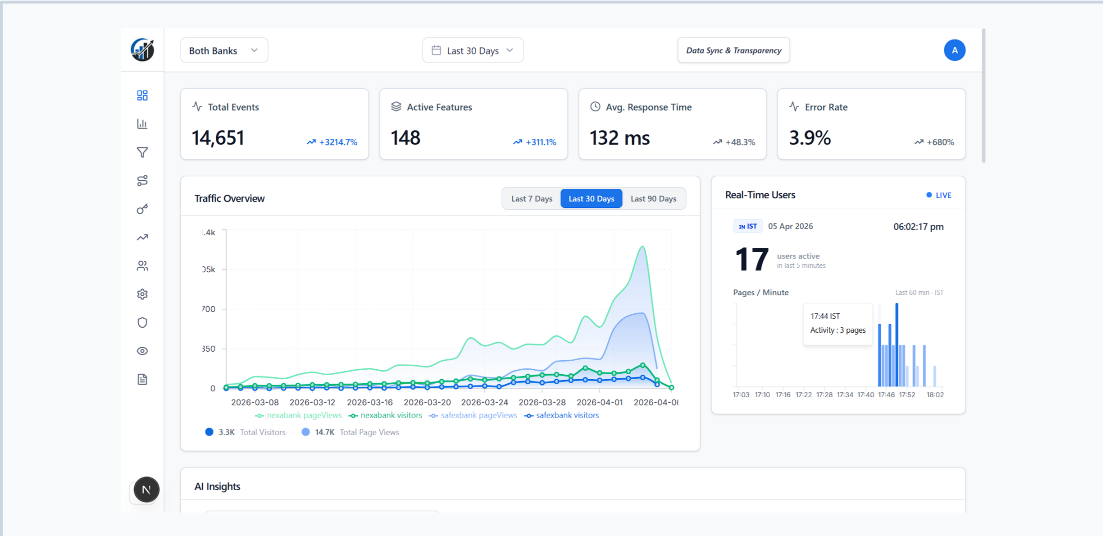

Figure 1. Dashboard Overview

Feature Analytics

Feature-level analysis combines trend charts, top interaction rankings, and heatmap intensity to reveal product adoption quality and behavior concentration.

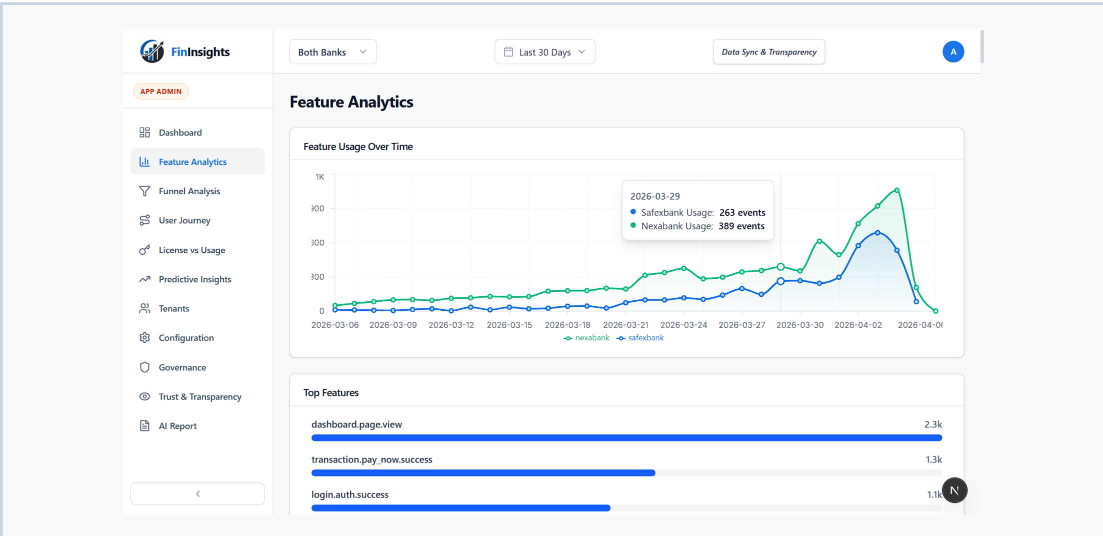

Figure 2. Feature Analytics

Funnel Analysis

This view quantifies stage-wise conversion, highlights leak severity, and supports targeted optimization where user drop-offs have the highest business impact.

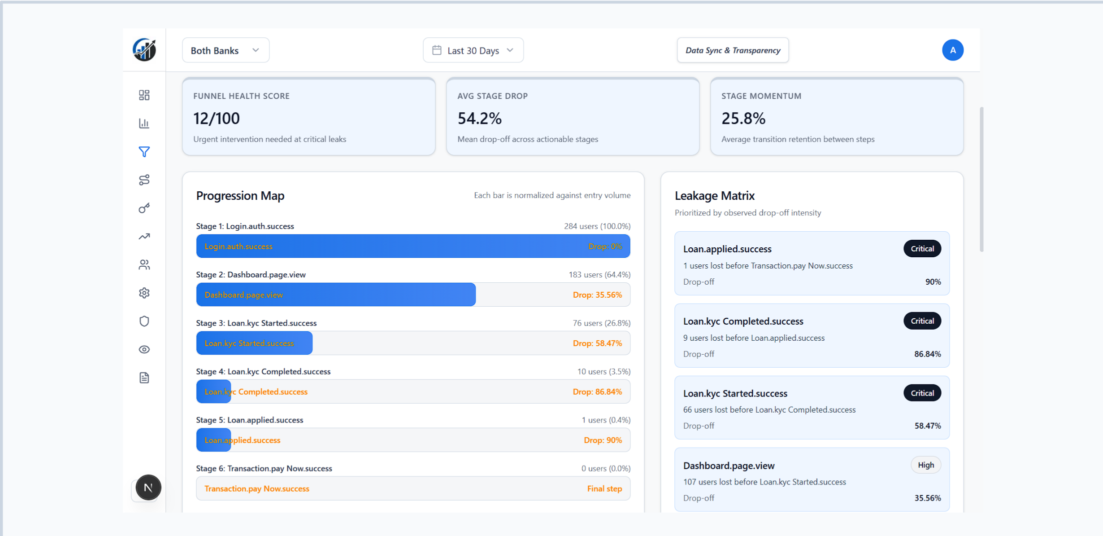

Figure 3. Funnel Analysis

License Usage Intelligence

License analytics maps entitled capabilities against actual usage, helping teams identify waste, underutilized premium modules, and growth opportunities.

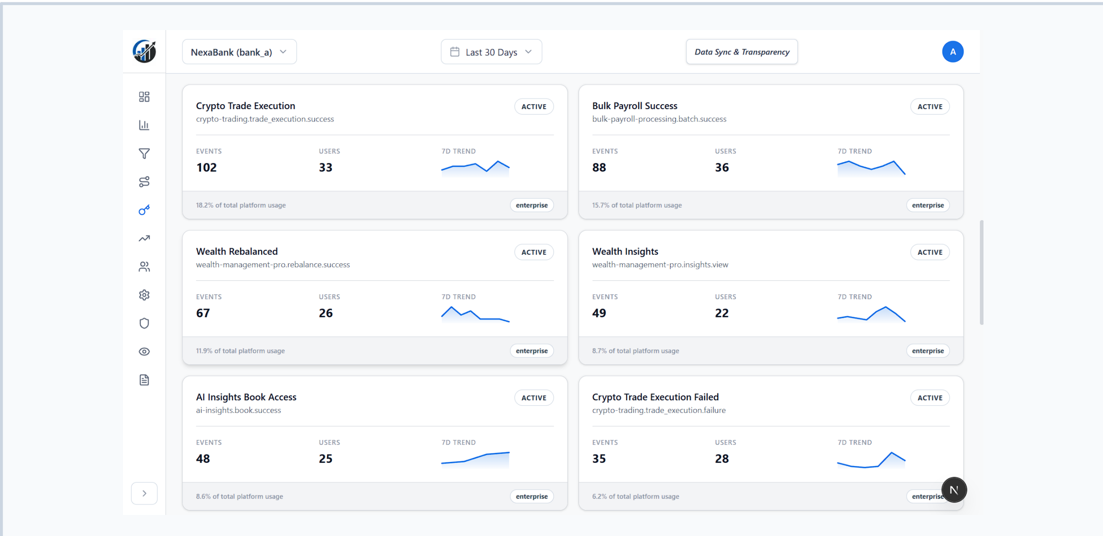

Figure 4. License Usage Intelligence

Multi-Tenant Comparison

Cross-tenant benchmarking for nexabank and safexbank enables direct comparison of engagement, feature usage, and conversion trajectories.

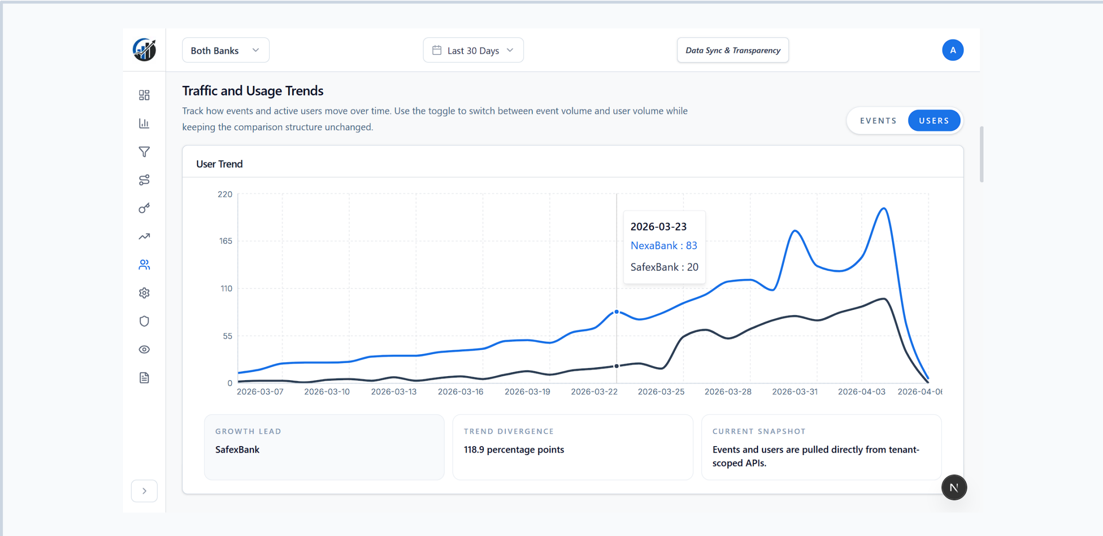

Figure 5. Multi-Tenant Comparison

Predictive Analytics

Predictive surfaces estimate adoption direction and future movement, allowing product teams to act before lagging indicators manifest in production.

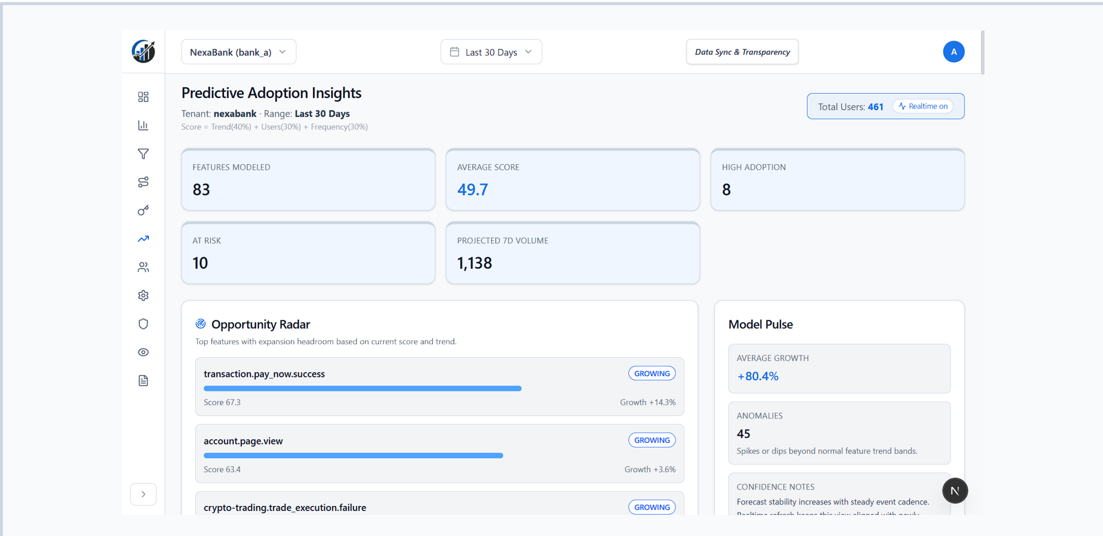

Figure 6. Predictive Analytics

Transparency and Governance

Transparency pages support governance by exposing controlled data surfaces and clear operational visibility across analytical workflows.

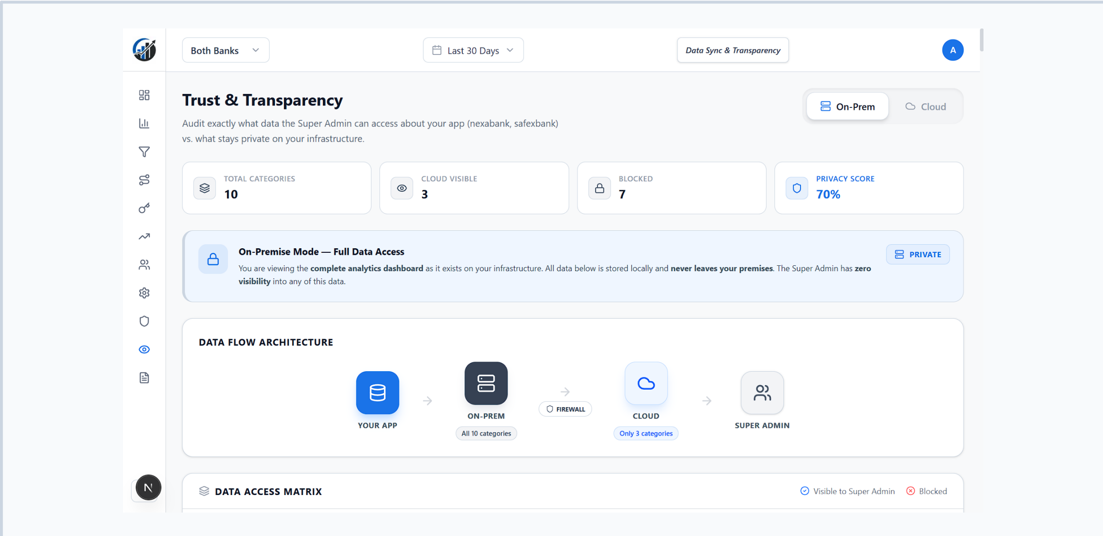

Figure 7. Transparency and Governance

AI Summary Report

The AI report converts telemetry and aggregated metrics into executive narratives, accelerating communication between technical teams and leadership stakeholders.

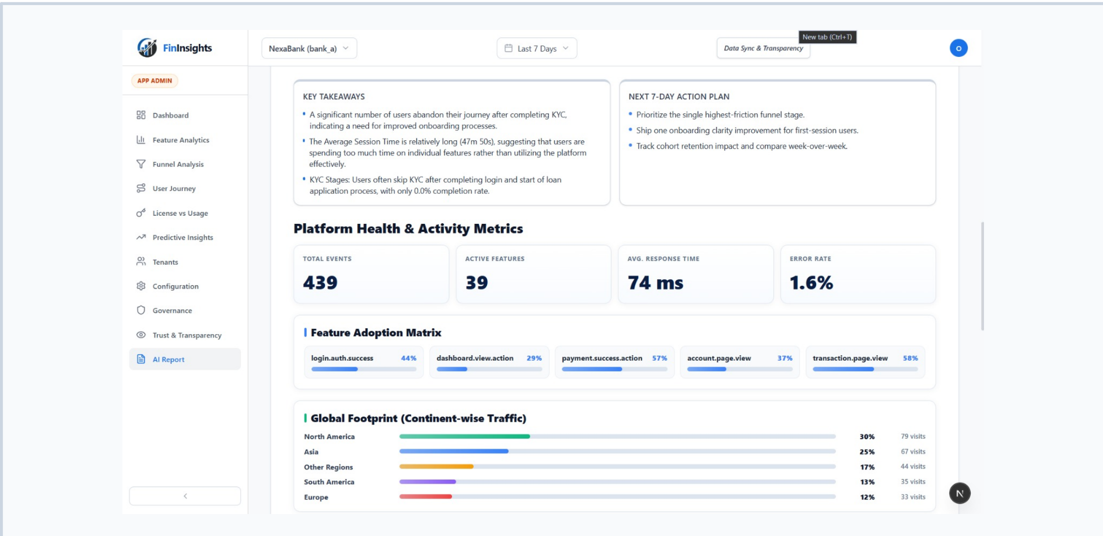

Figure 8. AI Summary Report

Conclusion

FinInsights is a real-time decision engine for modern fintech operations. By combining the KFC analytics stack with strong multi-tenant governance and local AI, it provides a foundation that is both secure for banking workloads and fast enough for continuous product iteration.

Project Status: Production-Ready (Development Tier)

Note: NexaBank is configured with two tenant contexts (nexabank and safexbank), enabling clear side-by-side multi-tenant comparison within FinInsights.